Q3LOCK Renyi History, OS Gap, and Low-Doublet Route Split

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This gate-level checkpoint records EXP-000805 and advances reusable result R-167 to additive version v1.8 without allocating a new result number. Its stable identifier remains PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT. The task is T-054, the claim context is C6-SPACETIME-SIGNATURE, the tier is T0, and claim_bearing is false.

Five narrow subgates close in this version:

1. exact inductive-exhaustion compatibility at every fixed finite Trotter level;
2. reduction of the two-orientation history tail to a uniform sandwiched-Renyi estimate;
3. equivalence of a zero-temperature phasewise GNS gap, coercivity, and OS temporal mass;
4. the exact minimum of the isolated one-site Q3 instanton action;
5. an exact lower bound for the gap of a conditional low-doublet Ising reference Hamiltonian.

Two mathematical parent gates remain open. The first still requires an actual Q3LOCK two-orientation history estimate, growing-stage $n \longrightarrow \infty$ Cauchy control, and a common all-exhaustion continuous dynamics. The second still requires actual broken-phase selection and volume-uniform GNS coercivity for the exact Q3LOCK Hamiltonian. The prospective gate PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE also remains open.

This checkpoint therefore does not prove a thermodynamic split dynamics, a phase-KMS quotient, beta-infinity phase selection, an actual Q3LOCK sandwiched-Renyi bound, an actual broken-sector temporal mass or spectral gap, regulator removal, continuum, a same-Hamiltonian physical-empty comparison, prospective validation, C6, CP1, physical Sector A, or Pre-A closure.

2 Provenance and gate ledger

The direct parents are EXP-000782, EXP-000789, and EXP-000804; the present decision is EXP-000805. The five closed subgates are exactly

1. PA-CP1-ST8-Q3LOCK-FIXED-TROTTER-LOCAL-STRICT-INDUCTIVE-EXHAUSTION-COMPATIBILITY;
2. PA-CP1-ST8-Q3LOCK-SANDWICHED-RENYI-TO-TWO-ORIENTATION-HISTORY-TAIL-CORRIDOR-REDUCTION;
3. PA-CP1-ST8-Q3LOCK-PHASEWISE-GNS-GAP-OS-TEMPORAL-MASS-EQUIVALENCE;
4. PA-CP1-ST8-Q3LOCK-ONE-SITE-Q3-INSTANTON-ACTION-MINIMUM;

5. PA-CP1-ST8-Q3LOCK-CONDITIONAL-DOUBLET-ISING-REFERENCE-GAP.

The two open mathematical successors are

1. PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA;
2. PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY.

The old combined gate PA-CP1-ST8-Q3LOCK-QUASI-LOCAL-RAW-OSCILLATOR-ALL-EXHAUSTION-COMMON-ALPHA-AND-BROKEN-GNS-GAP is historically open and superseded as the active successor; it is not a closed theorem. The following v1.7 route obligations are retained:

1. PA-CP1-ST8-Q3LOCK-ALL-EXHAUSTION-MIXTURE-L2-LOCALITY-AND-BETA-INDEPENDENT-CSTAR-DYNAMICS;
2. PA-CP1-ST8-Q3LOCK-HAMILTONIAN-THERMODYNAMIC-IDENTIFICATION-IN-CANONICAL-OS-MIXTURE;
3. PA-CP1-ST8-Q3LOCK-PROJECTED-DUHAMEL-MODULAR-C1-MULTIPLIER-LOCALITY.

Two scoped failures are new:

1. NG-2026-08-11-PRE-A-ST8-Q3LOCK-ENERGY-FORM-ENTROPY-FINITE-MOMENT-AUTOMATIC-SANDWICHED-RENYI-UPGRADE;
2. NG-2026-08-11-PRE-A-ST8-Q3LOCK-DIRECT-YAROTSKY-TWO-PHASE-GAP-IMPORT.

The prior failures NG-2026-08-11-PRE-A-ST8-Q3LOCK-ENTROPY-FINITE-MOMENT-DYNAMIC-GAUSSIAN-TAIL-INFERENCE and NG-2026-08-11-PRE-A-ST8-Q3LOCK-ORDERED-GROUND-DOUBLETS-AUTOMATIC-GNS-GAP remain applicable and are not promoted beyond their registered scopes.

3 Exact fixed-Trotter exhaustion compatibility

For a finite lattice set X, let

$$\mathcal{A}_X = B\Big(L^2(\mathbb{R}^{8|X|})\Big) \quad \text{and} \quad \mathcal{A}_X \longrightarrow \mathcal{A}_Y, \quad A \longmapsto A \otimes 1, \tag{3.1}$$

whenever $X \subset Y$. An exact onsite subflow preserves support. Every cross-bond generator

$$v_{xy} = -c\, q_x \cdot q_y \tag{3.2}$$

is a multiplication polynomial. All such generators commute exactly, including generators on bonds with one common endpoint. Factors on bonds not incident on the current support commute with the observable and with every retained bond factor. Consequently one exact all-bond layer maps

$$\mathcal{A}_X \longrightarrow \mathcal{A}_{N_1(X)} \tag{3.3}$$

and agrees in every finite ambient volume containing $N_1(X)$.

More generally, a fixed split-product word containing b bond layers lies in $\mathcal{A}_{N_b(X)}$ and is exactly independent of every ambient volume containing that halo. In particular, for fixed n,

$$\Theta_{t,n} = (\sigma_{t/n}\beta_{t/n})^n, \qquad \Theta_{t,n} : \mathcal{A}_X \longrightarrow \mathcal{A}_{N_n(X)}, \tag{3.4}$$

and the reverse-sign inverse word is

$$\Theta_{t,n}^{-1} = (\beta_{-t/n}\sigma_{-t/n})^n. \tag{3.5}$$

Adjoints obey the same support statement. Local Weyl and basic-resolvent labels may be inserted as bounded finite-region multiplier inputs. Nothing here proves that the outputs preserve one inductive local Weyl or resolvent C^*-algebra.

3.1 Exact constructive support proxy

The primary and independent engines verify the support mechanism without using a symbolic support flag. For the cubic lattice, the first four ℓ^1-ball cardinalities around one site are

$$|N_0|, |N_1|, |N_2|, |N_3| = 1, 7, 25, 63. \tag{3.6}$$

A separate one-dimensional rational proxy starts from $q_0 = 1$, all other $q_x, p_x = 0$, and uses

$$\kappa = \frac{10}{21}, \qquad p_x \longmapsto p_x + \kappa(q_{x-1} + q_{x+1}), \qquad (q_x, p_x) \longmapsto (p_x, -q_x). \tag{3.7}$$

After three forward layers the exact nonzero word is

$$\begin{array}{c|ccccccc} x & -3 & -2 & -1 & 0 & 1 & 2 & 3 \\ \hline q_x & \frac{1000}{9261} & 0 & -\frac{1940}{3087} & 0 & -\frac{1940}{3087} & 0 & \frac{1000}{9261} \\ p_x & 0 & -\frac{100}{441} & 0 & \frac{241}{441} & 0 & -\frac{100}{441} & 0. \end{array} \tag{3.8}$$

Ambient radii three and four give exactly (3.8) at every prefix, the reverse word recovers the seed exactly, and ambient radius two gives a different word. Thus the outer halo is both sufficient and detected by a nonzero coefficient. This proxy checks the exact support and inverse bookkeeping; it is not a substitute for the nonlinear Q3 onsite flow.

Equations (3.1)–(3.8) are fixed-finite-word statements. The support radius grows with n, so they prove no $n \longrightarrow \infty$ Cauchy estimate, continuous group, common generator, global strict topology, or phase-KMS quotient.

4 Sandwiched-Renyi history domination

Let ρ be a faithful finite Gibbs density, let P be a unitary partial split history, let $\alpha > 1$, and set

$$\vartheta = \frac{\alpha - 1}{\alpha}, \qquad \widetilde{Q}_\alpha(P\rho P^* \| \rho) = \left\| \rho^{-\vartheta/2} P \rho^{1/2} \right\|_{2\alpha}^{2\alpha}. \tag{4.1}$$

For every projection E, noncommutative Holder, or equivalently binary measurement data processing, gives

$$\operatorname{Tr}(P\rho P^* E) \le \widetilde{Q}_\alpha(P\rho P^* \| \rho)^{1/\alpha} \operatorname{Tr}(\rho E)^\vartheta. \tag{4.2}$$

Suppose that both orientations, uniformly over the finite volume, compact source, mesh, partial history, and $|t| \le T$, satisfy

$$\begin{aligned} \widetilde{Q}_\alpha(P\rho P^* \| \rho) &\le Q_{\alpha,T}, \\ \widetilde{Q}_\alpha(P^* \rho P \| \rho) &\le Q_{\alpha,T}. \end{aligned} \tag{4.3}$$

Then the required orientation sum has the exact factor two:

$$\rho(P^*EP) + \rho(PEP^*) \le 2Q_{\alpha,T}^{1/\alpha}\rho(E)^{\vartheta}. \tag{4.4}$$

This is a sufficient reduction. The smaller model-specific target is (4.4) and its coordinate-polynomial weighted form on the actual cutoff events; the full uniform Renyi hypothesis is not asserted to be necessary.

4.1 Noncommuting orientation oracle

The exact two-level oracle is deliberately noncommuting. Put

$$\rho = \begin{pmatrix} 4/5 & 0 \\ 0 & 1/5 \end{pmatrix}, \qquad \sigma_{\pm} = \begin{pmatrix} 52/125 & \pm 36/125 \\ \pm 36/125 & 73/125 \end{pmatrix}. \tag{4.5}$$

For $\alpha = 2$,

$$\widetilde{Q}_2(\sigma_+\|\rho) = \widetilde{Q}_2(\sigma_-\|\rho) = \frac{7301}{3125}, \qquad Q_2^{\mathrm{Petz}}(\sigma_\pm\|\rho) = \frac{61}{25}. \tag{4.6}$$

For $E_+ = |+\rangle\langle +|$,

$$\rho(E_+) = \frac{1}{2}, \qquad \sigma_+(E_+) = \frac{197}{250}, \qquad \sigma_-(E_+) = \frac{53}{250}. \tag{4.7}$$

Thus the two orientations are distinct even though their sandwiched costs agree, and the Petz and sandwiched quantities are not interchangeable. Equations (4.5)–(4.7) protect both the orientation bookkeeping and the choice of Renyi functional.

5 Weighted coordinate tail and the cutoff corridor

For a finite coordinate set S, define

$$X_S = \max_{x\in S} |q_x|, \qquad E_{S,L} = 1_{\{X_S > L\}}. \tag{5.1}$$

Assume the registered reference tail

$$\rho(E_{S,L}) \le M_a|S|e^{-aL^2}. \tag{5.2}$$

With $b = \vartheta a$, applying (4.2) at every radius and integrating the tail yields, for either history orientation,

$$\begin{aligned} \mathrm{Tr}(P\rho P^* X_S^4 E_{S,L}) \le\ & Q_{\alpha,T}^{1/\alpha}(M_a|S|)^{\vartheta} e^{-bL^2} \\ & \times \left(L^4 + \frac{2L^2}{b} + \frac{2}{b^2}\right). \end{aligned} \tag{5.3}$$

For one cutoff bond,

$$|w_{xy,L}|^2 \le 4c^2 X_{\{x,y\}}^4 E_{\{x,y\},L}. \tag{5.4}$$

If at most m edges occur, Cauchy–Schwarz and the two orientations give the explicit bound

$$\begin{aligned} & \rho\left(P^*|\sum_e w_{e,L}|^2 P\right) + \rho\left(P|\sum_e w_{e,L}|^2 P^*\right) \\ & \quad \le 8c^2m^2 Q_{\alpha,T}^{1/\alpha}(M_a|S|)^{\vartheta} e^{-bL^2}\left(L^4 + \frac{2L^2}{b} + \frac{2}{b^2}\right). \end{aligned} \tag{5.5}$$

The exact coefficient oracle takes

$$\alpha = 2, \quad \vartheta = \frac{1}{2}, \quad a = 12, \quad b = 6, \quad L = 2, \quad Q = 9, \quad M_a|S| = 16, \quad c = \frac{3}{5}, \quad m = 3. \tag{5.6}$$

Then

$$L^4 + \frac{2L^2}{b} + \frac{2}{b^2} = \frac{313}{18}, \tag{5.7}$$

the one-orientation prefactor before this polynomial is 3888/25, the two-orientation prefactor is 7776/25, and (5.5) becomes

$$\frac{135216}{25} e^{-24}. \tag{5.8}$$

The bounded-cutoff propagation term has the safe form

$$8\sqrt{2}\,|X|\,\|A\| e^{\nu_L T} \frac{(\nu_L T)^R}{R!}, \qquad \nu_L \le \frac{96cL^2}{\hbar}. \tag{5.9}$$

With $L = R^\gamma$ and $0 < \gamma < 1/2$, its factorial part vanishes. If the sandwiched divergence satisfies

$$\widetilde{D}_\alpha(P\rho P^* \| \rho) \le dL^2 + o(L^2), \tag{5.10}$$

then (5.3) has effective exponential rate $\vartheta(a-d)$. A history multiplier $e^{\kappa_T L^2}$ paid in a squared seminorm is absorbed when

$$\vartheta(a-d) > \kappa_T, \tag{5.11}$$

whereas the safe condition after taking the square root is

$$\vartheta(a-d) > 2\kappa_T. \tag{5.12}$$

For the corridor oracle

$$\alpha = 2, \quad a = 14, \quad d = 1, \quad \kappa_T = 3, \quad \gamma = \frac{1}{3}, \tag{5.13}$$

the effective rate is 13/2, both (5.11) and (5.12) hold, and the factorial exponent $2\gamma - 1$ is $-1/3$. These exact constants close the arithmetic conditional on (4.3); they do not prove (4.3) for Q3LOCK.

6 Exact energy-form and finite-moment no-go

The current entropy, energy-form, and finite-moment inputs do not automatically imply the Renyi hypothesis. An exact rational family makes the separation explicit. For integers $n \ge 2$ and $m \ge 3$, set

$$R_n = 2^{n^4}, \qquad p_{0,n} = \frac{R_n}{R_n + 1}, \qquad p_{1,n} = \frac{1}{R_n + 1}, \qquad \delta_n = p_{0,n} - p_{1,n}. \tag{6.1}$$

Let

$$d_{n,m} = 4n^{2m} + 1, \qquad c_{n,m} = \frac{4n^{2m} - 1}{d_{n,m}}, \qquad s_{n,m} = \frac{4n^m}{d_{n,m}}, \tag{6.2}$$

so that $c_{n,m}^2 + s_{n,m}^2 = 1$. Rotate $\rho_n = \mathrm{diag}(p_{0,n}, p_{1,n})$ by either sign of the real rotation with cosine $c_{n,m}$ and sine $s_{n,m}$. For the excited projection P_1,

$$q_{n,m} = \mathrm{Tr}(U_\pm \rho_n U_\pm^* P_1) = p_{1,n} + \delta_n s_{n,m}^2. \tag{6.3}$$

With

$$K_n = \operatorname{diag}(1, 1+n^4), \qquad X_n = nP_1, \tag{6.4}$$

the exact relative entropy and energy excess are

$$\begin{aligned} D(U_\pm \rho_n U_\pm^* \| \rho_n) &= \delta_n s_{n,m}^2 n^4 \log 2, \\ \operatorname{Tr}(U_\pm \rho_n U_\pm^* K_n) - \operatorname{Tr}(\rho_n K_n) &= \delta_n s_{n,m}^2 n^4. \end{aligned} \tag{6.5}$$

They vanish as $O(n^{4-2m})$. The generalized relative matrix has determinant one and trace strictly below $5/2$, hence

$$\frac{1}{2} K_n \le U_\pm K_n U_\pm^* \le 2K_n. \tag{6.6}$$

For every $1 \le r \le 2m$,

$$\operatorname{Tr}(U_\pm \rho_n U_\pm^* X_n^r) = n^r q_{n,m} \tag{6.7}$$

is uniformly bounded. By increasing m, this covers any preregistered finite list of polynomial moments. The reference also has every fixed Gaussian coefficient:

$$\operatorname{Tr}(\rho_n e^{aX_n^2}) = \frac{R_n + e^{an^2}}{R_n + 1}, \tag{6.8}$$

which is uniformly bounded for each fixed a.

Nevertheless,

$$q_{n,m} \ge \frac{8}{25n^{2m}}, \tag{6.9}$$

and binary measurement data processing gives, for every fixed $\alpha > 1$,

$$\widetilde{D}_\alpha(U_\pm \rho_n U_\pm^* \| \rho_n) \ge n^4 \log 2 - \frac{2m\alpha}{\alpha - 1} \log n + O(1) \longrightarrow \infty. \tag{6.10}$$

The compact exact oracle at $n = 2, m = 3$ is

$$\begin{array}{c|c} p_0 & 65536/65537 \\ p_1 & 1/65537 \\ c_{2,3} & 255/257 \\ s_{2,3} & 32/257 \\ q_{2,3} & 261377/16843009 \\ \operatorname{Tr} G & 2507810/1122833 \\ Q_2^{\text{measured}} & 1106449/66049 \\ \widetilde{Q}_2 & 106339353113/4328653313 \\ \widetilde{Q}_2 - Q_2^{\text{measured}} & 33826005000/4328653313 \\ q_{2,3}^2/p_1 & 68317936129/4328653313. \end{array} \tag{6.11}$$

This proves only failure of the automatic implication. The auxiliary rotations are not Q3LOCK histories, so (6.1)–(6.11) neither refute an actual model-specific Q3LOCK quasi-invariance theorem nor prove dynamics nonexistence.

7 Zero-temperature OS mass and GNS gap equivalence

Let $(\pi_\omega, \mathcal{H}_\omega, \Omega_\omega)$ be one selected zero-temperature phase representation with

$$H_\omega \geq 0, \qquad H_\omega \Omega_\omega = 0, \qquad \ker H_\omega = \mathbb{C}\Omega_\omega. \tag{7.1}$$

For an invariant unital observable core, put

$$\xi_A = \pi_\omega(A - \omega(A)1)\Omega_\omega. \tag{7.2}$$

Assume that these centered vectors form a form core for $H_\omega^{1/2}$ on $\Omega_\omega^\perp$. Define physical imaginary time

$$\begin{aligned} G_A(\tau) &= \langle \xi_A, e^{-\tau H_\omega/\hbar} \xi_A \rangle \\ &= \int_{(0,\infty)} e^{-\tau E/\hbar} \, d\mu_A(E). \end{aligned} \tag{7.3}$$

For every $\Delta > 0$, the following are equivalent:

1. $\sigma(H_\omega) \subset \{0\} \cup [\Delta, \infty)$;
2. for every core observable,

$$-i\hbar \, \omega(A^* \delta(A)) \geq \Delta\{\omega(A^* A) - |\omega(A)|^2\}; \tag{7.4}$$

3. for every core observable and $\tau \geq 0$,

$$G_A(\tau) \leq e^{-\Delta \tau/\hbar} G_A(0). \tag{7.5}$$

Here $\delta(A) = (i/\hbar)[H_\omega, A]$ on the core, and

$$-\hbar G_A'(0+) = \langle \xi_A, H_\omega \xi_A \rangle = -i\hbar \, \omega(A^* \delta(A)). \tag{7.6}$$

The spectral theorem proves the forward implications; the form-core hypothesis promotes the converse coercivity to the whole vacuum complement. A weaker large-time test is also exact: if every core observable has a finite C_A and there is one common $m > 0$ such that, for all sufficiently large τ,

$$G_A(\tau) \leq C_A e^{-m\tau}, \tag{7.7}$$

then no core spectral measure has support below $\hbar m$, and the form-core hypothesis gives

$$\Delta \geq \hbar m. \tag{7.8}$$

This theorem identifies the precise spectral target. A fixed-beta periodic OS/KMS reconstruction supplies a thermal Liouvillean, not automatically the positive vacuum Hamiltonian in (7.1). Q3LOCK still needs beta-infinity phase selection and one phasewise, beta-uniform half-line rate.

8 Exact one-site Q3 instanton action

For the shifted one-site potential, write

$$U(q) = \frac{g}{4}\sum_{e=1}^{8}(q_e^2 - v^2)^2 + \frac{\lambda}{4}\sum_{\{e,f\}\in E(Q_3)}(q_e - q_f)^2(q_e^2 + q_f^2), \tag{8.1}$$

where $\chi > 0$, $v > 0$, $g > 0$, and $\lambda \geq 0$. For every absolutely continuous finite-action H^1_{loc} heteroclinic with $\dot{q} \in L^2$, $q(-\infty) = -v\mathbf{1}$, and $q(+\infty) = v\mathbf{1}$,

$$I[q] = \int_{\mathbb{R}}\left(\frac{\chi}{2}|\dot{q}|^2 + U(q)\right) dt \geq S_{\mathrm{inst}} := \frac{16\sqrt{2}}{3}v^3\sqrt{\chi g}. \tag{8.2}$$

Indeed, discarding the nonnegative Q3 coupling and applying the pointwise arithmetic-geometric mean inequality gives

$$\frac{\chi}{2}|\dot{q}|^2 + \frac{g}{4}\sum_e(q_e^2 - v^2)^2 \geq \sqrt{\frac{\chi g}{2}}\sum_e|\dot{q}_e|\,|q_e^2 - v^2|. \tag{8.3}$$

Each of the eight endpoint variations contributes at least $4v^3/3$. Equality is attained by the locked path

$$q_e(t) = v\tanh\left(v\sqrt{\frac{g}{2\chi}}(t - t_0)\right). \tag{8.4}$$

When $\lambda > 0$, equality in the Q3 term locks the eight centers, so the minimizer is unique up to common time translation. When $\lambda = 0$, the eight scalar centers may translate independently.

For the exact verifier fixture

$$\chi = 1, \qquad g = 2, \qquad v = \frac{3}{2}, \qquad \lambda = \frac{5}{7}, \tag{8.5}$$

the squared action is 1296 and

$$S_{\mathrm{inst}} = 36. \tag{8.6}$$

This is an action minimum, not an eigenvalue-splitting theorem. It proves neither a tunnelling prefactor nor a volume-uniform sector gap.

9 Conditional low-doublet Ising reference

Assume that the exact one-site Hamiltonian has simple even and odd $\mathrm{Aut}(Q_3)$ singlets ϕ_0, ϕ_1 with

$$\epsilon_0 < \epsilon_1 < \epsilon_2. \tag{9.1}$$

Define

$$\begin{aligned} P &= |\phi_0\rangle\langle\phi_0| + |\phi_1\rangle\langle\phi_1|, \\ s &= |\phi_0\rangle\langle\phi_1| + |\phi_1\rangle\langle\phi_0|, \\ \delta_1 &= \epsilon_1 - \epsilon_0, \\ k &= h_{\mathrm{site}} - \epsilon_0 - \delta_1|\phi_1\rangle\langle\phi_1|. \end{aligned} \tag{9.2}$$

If

$$\Gamma = \inf \sigma(h_{\mathrm{site}}|_{P^\perp}) - \epsilon_0 > 0, \qquad k \geq \Gamma(1-P), \tag{9.3}$$

choose phases so that, independently of the Q3 coordinate e,

$$m = \langle \phi_0, q_e \phi_1 \rangle > 0, \qquad R_e = q_e - ms. \tag{9.4}$$

Parity and symmetry give $PR_eP = 0$. On a connected finite lattice graph G, define

$$H_{\mathrm{ref}} = \sum_x k_x + \frac{cm^2}{2} \sum_{\langle xy \rangle, e} (s_x - s_y)^2, \qquad J = 8cm^2. \tag{9.5}$$

Its kernel is exactly the span of the two product vectors

$$\bigotimes_x \frac{\phi_0 + \phi_1}{\sqrt{2}}, \qquad \bigotimes_x \frac{\phi_0 - \phi_1}{\sqrt{2}}. \tag{9.6}$$

If $\kappa(G)$ is the edge connectivity, then

$$\mathrm{gap}(H_{\mathrm{ref}}) \geq \min\{\Gamma, 2J\kappa(G)\}. \tag{9.7}$$

A state outside the doublet sector pays at least Γ. Within the doublet sector, every disagreeing bond pays $2J$, and every nonconstant configuration cuts at least $\kappa(G)$ bonds. For the standard periodic cubic bond multigraph of side at least two, $\kappa(G) = 6$ and the bound is $\min\{\Gamma, 12J\}$.

The exact full-minus-reference decomposition is

$$\begin{aligned} H - \epsilon_0|\Lambda| = H_{\mathrm{ref}} &+ \delta_1 \sum_x P_{1,x} \\ &+ \frac{c}{2} \sum_{\langle xy \rangle, e} \Big[m\{s_x - s_y, R_{x,e} - R_{y,e}\} + (R_{x,e} - R_{y,e})^2 \Big]. \end{aligned} \tag{9.8}$$

The verifier fixture uses

$$\delta_1 = \frac{1}{10}, \quad \Gamma = 5, \quad m = \frac{1}{2}, \quad c = \frac{3}{5}, \quad J = \frac{6}{5}, \quad \kappa = 6. \tag{9.9}$$

The low-sector excitation is $72/5$, and the certified reference lower bound is therefore 5. An exact three-level matrix fixture independently checks the bond expansion in (9.8).

The reference-gap certificate and the transfer to the exact model have different assumptions. The reference certificate first needs rigorous enclosures for δ_1, Γ, m to validate (9.3)–(9.7). Even after that reference gap is available, transfer to the exact Q3LOCK Hamiltonian still needs relative-form estimates for every R_e and a volume-uniform remainder-smallness, perturbation, or block-diagonalization theorem with controlled truncation removal. Equation (9.7) is a gap lower bound for a conditional reference Hamiltonian, not the actual Q3LOCK broken-sector GNS gap.

10 Boundary of the Yarotsky imports

The direct two-phase quantum Pirogov–Sinai import does not meet its load-bearing hypotheses. In the form audited here, it requires a finite-dimensional onsite reference, two exact Hilbert-space product zero vectors minimizing the local reference blocks, a local gap and Peierls condition, a nonzero first-order splitting vector, and a sufficiently small relative perturbation. The exact Q3 onsite space is $L^2(\mathbb{R}^8)$. Its classical point minima are not Hilbert vectors, its finite quantum Schrodinger ground vector is unique and positive, and neither the controlled doublet embedding nor the required smallness of (9.8) has been proved.

The distinct single-phase relative-form theorem permits an infinite-dimensional onsite Hilbert space. For every $\eta > 0$, the Q3 bond obeys a bound of the form

$$
\begin{aligned}
B_{xy} \le \eta[(h_x - \epsilon_0) + (h_y - \epsilon_0)] \\
+ \left(2\eta\epsilon_0 + 4\eta g v^4 + \frac{32c^2}{\eta g}\right) 1.
\end{aligned} \tag{10.1}
$$

The scalar residual used in this estimate is the exact sum of squares

$$
\frac{\eta g}{8}(q^2 - 2v^2)^2 + \frac{\eta g}{8}\left(q^2 - \frac{4c}{\eta g}\right)^2. \tag{10.2}
$$

This theorem can yield an existential unique gapped weak-coupling phase near a simple product reference. It does not yield the desired ordered two-phase conclusion.

The registered infrared condition

$$
c > c_{\mathrm{IR}} := \frac{\hbar^2 J_3^2}{8\chi\theta_Q^2} \tag{10.3}
$$

is only a lower bound on c. It supplies no upper perturbative radius and none of δ_1, Γ, m or the R_e constants required by (9.8). Therefore the direct two-phase import fails by a hypothesis mismatch. This is not a no-go for a future controlled low-doublet theorem and not a counterexample to an actual sector gap.

The primary sources for this distinction are Yarotsky's https://arxiv.org/abs/math-ph/0412040 single-phase relative-perturbation theorem and the two-phase quantum Pirogov–Sinai statement at https://www.mathnet.ru/eng/rm1728.

11 Exact dependency boundary for Sector A and Pre-A

The dynamics parent is now recorded as

> **OPEN, with fixed-Trotter exhaustion compatibility and a conditional sandwiched-Renyi tail reduction closed on 2026-08-11.**

The spectral parent is now recorded as

> **OPEN, with the zero-temperature OS temporal-mass equivalence, one-site instanton minimum, and conditional reference gap isolated on 2026-08-11.**

The active `A5-SECTOR-A-SYNTHESIS` remains the published `T6` conditional composition under its seven registered hypotheses. This checkpoint does not alter that card and does not promote it to a physical or full-Class-II Sector-A theorem.

Physical Pre-A still requires target-algebra beta-infinity ground-state selection, actual broken-sector coercivity, enlarged-counterterm continuum control, a same-Hamiltonian physical-empty comparison, and prospective validation. The prospective gate requires a genuinely future or blind target together with a microscopic observable map frozen before disclosure. The present Round-1 record contains no admitted microscopic survivor. Repository-internal proof work cannot retroactively create the required time ordering. Pre-A therefore remains open.

12 Devil's-advocate review

1. **UPHELD as an overreach:** fixed-n compatibility does not imply the split limit. The halo grows with n, and no growing-stage Cauchy bound has been proved.

2. **DISMISSED:** the orientation theorem does not confuse Petz and sandwiched Renyi quantities. Equations (4.5)–(4.7) make them numerically different and check both history orientations.

3. **DISMISSED:** v1.7 entropy and energy comparability do not imply the required Renyi bound. Section 6 preserves a factor-two energy-form comparison and arbitrary preregistered finite moments while every fixed $\alpha > 1$ sandwiched divergence diverges.

4. **UPHELD as an overreach:** the exact no-go family is auxiliary, not a Q3LOCK history. It rejects only the automatic inference.

5. **UPHELD as an overreach:** a fixed-beta OS decay statement is not the vacuum gap. Section 7 requires a selected zero-temperature representation and a positive implementing Hamiltonian.

6. **UPHELD as an overreach:** the instanton action is not a tunnelling splitting or prefactor theorem.

7. **DISMISSED:** the conditional Ising reference is not silently identified with the exact Q3LOCK Hamiltonian. Equation (9.8) displays the unbounded remainder and the one-site splitting still to be controlled.

8. **DISMISSED:** the Yarotsky audit does not assert that all Yarotsky theorems require finite onsite dimension. The infinite-dimensional single-phase theorem is separated explicitly from the two-phase import.

9. **DISMISSED:** five narrow closures do not close physical Sector A or Pre-A. The two mathematical parents and the prospective holdout remain open.

13 Reproduction and evidence contract

The formal owner is the manifest `strategy/pre-a-cp1-st8-q3lock-renyi-history-os-gap-reduction-route-split-manifest.json`; the analytic authority is `strategy/pre-a-cp1-st8-q3lock-renyi-history-os-gap-reduction-route-split-certificate-260811.md`. The exact primary, non-importing independent, and integrated engines are

1. `codes/foundations/pre_a_cp1_st8_q3lock_renyi_history_os_gap_reduction_route_split.py`;

2. `codes/foundations/pre_a_cp1_st8_q3lock_renyi_history_os_gap_reduction_route_split_independent.py`;

3. `codes/foundations/pre_a_cp1_st8_q3lock_renyi_history_os_gap_reduction_route_split_verify.py`.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$stem = (
  'codes\foundations\pre_a_cp1_st8_q3lock_' +
  'renyi_history_os_gap_reduction_route_split'
)
```

```
& $py -X utf8 ($stem + '.py')
& $py -X utf8 ($stem + '_independent.py')
& $py -X utf8 ($stem + '_verify.py')
```

The expected proof-layer outputs are primary `PASS 127/127`, independent `PASS 115/115`, and integrated `PASS 197/197`. The independent engine does not import the primary engine and uses distinct exact OS and Ising-reference fixtures. The integrated layer launches both engines twice, checks deterministic canonical outputs, enforces its import firewall, cross-checks the exact constants and scope booleans, verifies stored-result freshness, audits all formal authorities, and governs the single gate-level source/PDF checkpoint.

This v0.7 file is the sole synthesis source for the present logical checkpoint. Its sibling PDF is governed by the manifest source and PDF hashes, page count, freshness, text extraction, and all-page visual review. No per-lemma or intermediate PDF belongs to this result.

14 Result footer

```
Result ID:               PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-
                         MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:         R-167 v1.8; no new result number.
Exploration:             EXP-000805.
Precise statement:       Fixed-Trotter compatibility; Renyi history-tail
                         reduction; zero-T OS mass/gap equivalence; exact
                         instanton minimum; conditional Ising reference gap.
Open parents:            All-exhaustion two-orientation history common alpha;
                         actual broken-sector GNS gap coercivity.
Prospective gate:        PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE
                         remains open.
Negative results:        No automatic Renyi upgrade from energy forms,
                         entropy and finite moments; no direct two-phase
                         Yarotsky import without its hypotheses.
Scope:                   T0 exact route reductions and counterfixture;
                         claim_bearing false.
Dependencies:            EXP-000782, EXP-000789, EXP-000804; R-167 v1.0-v1.7;
                         exact finite Q3 Hamiltonians and registered tails.
Evidence grade:          ANALYTIC + EXACT + primary + non-importing
                         independent + integrated verification.
Reproduction command:    Run the three commands in Section 13 in the listed
                         order, with the integrated verifier last.
Expected output:         Primary PASS 127/127; independent PASS 115/115;
                         integrated PASS 197/197.
Falsification gate:      Any support word, orientation, Petz/sandwiched,
                         edge coefficient, no-go rational, OS sign or rate,
                         instanton factor, min-cut gap, provenance, freshness,
                         hash, count or scope mismatch.
Tier before / after:     C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:  No n-to-infinity split limit, actual Q3LOCK Renyi
                         history estimate, all-exhaustion common alpha,
                         phase-KMS quotient, beta-infinity selection, actual
                         broken-sector temporal mass or gap, regulator
                         removal, continuum, physical-empty comparison,
                         prospective validation, C6, CP1, physical Sector A,
                         or Pre-A closure.
Next required action:    Prove uniform two-orientation cutoff domination and
                         growing-stage Cauchy completion; separately prove the
                         doublet remainder bound or a beta-uniform OS mass.
```